

UNIT 3 – EXERCISE 8

Define a Local Class in ABAP

ZCL_9309_LOCAL_CLASS

Status: COMPLETED ✓

1. Objective

Create a global ABAP class and define a local helper class LCL_CONNECTION inside it. The local class contains two instance attributes and one static/class attribute.

2. What is a Local Class?

A local class is defined inside the source code of another ABAP class. It is useful when the helper class is needed only by that program.

Global Class → ZCL_9309_LOCAL_CLASS

Local Class → LCL_CONNECTION

3. Task 1 – Create the Global Class

Class created: ZCL_9309_LOCAL_CLASS

Interface: IF_OO_ADT_CLASSRUN

1. File → New → ABAP Repository Object.
2. Select ABAP Class.
3. Enter the package and class name ZCL_9309_LOCAL_CLASS.
4. Add interface IF_OO_ADT_CLASSRUN.
5. Create the class and activate it.

4. Task 2 – Create the Local Class

6. Open the Local Types tab.
7. Type lcl and press Ctrl + Space.
8. Select lcl - class from code completion.
9. Rename the generated class to lcl_connection.
10. Remove CREATE PRIVATE.
11. Add the three required attributes.
12. Create the local class implementation.
13. Press Ctrl + F3 to activate.

5. Final Code

```
CLASS lcl_connection DEFINITION.  
    PUBLIC SECTION.  
  
        DATA carrier_id TYPE /dmo/carrier_id.  
        DATA connection_id TYPE /dmo/connection_id.  
        CLASS-DATA conn_counter TYPE i.  
  
    PROTECTED SECTION.  
    PRIVATE SECTION.  
ENDCLASS.
```

```
CLASS lcl_connection IMPLEMENTATION.  
ENDCLASS.
```

6. Attribute Explanation

Attribute	Declaration	Scope	Purpose
carrier_id	TYPE /dmo/carrier_id	Instance	Stores carrier/airline ID for an object.
connection_id	TYPE /dmo/connection_id	Instance	Stores connection ID for an object.
conn_counter	TYPE i	Static/Class	One shared counter for the whole class.

7. DATA vs CLASS-DATA

DATA creates an instance attribute: each object has its own value. CLASS-DATA creates a class/static attribute: all objects share the same value.

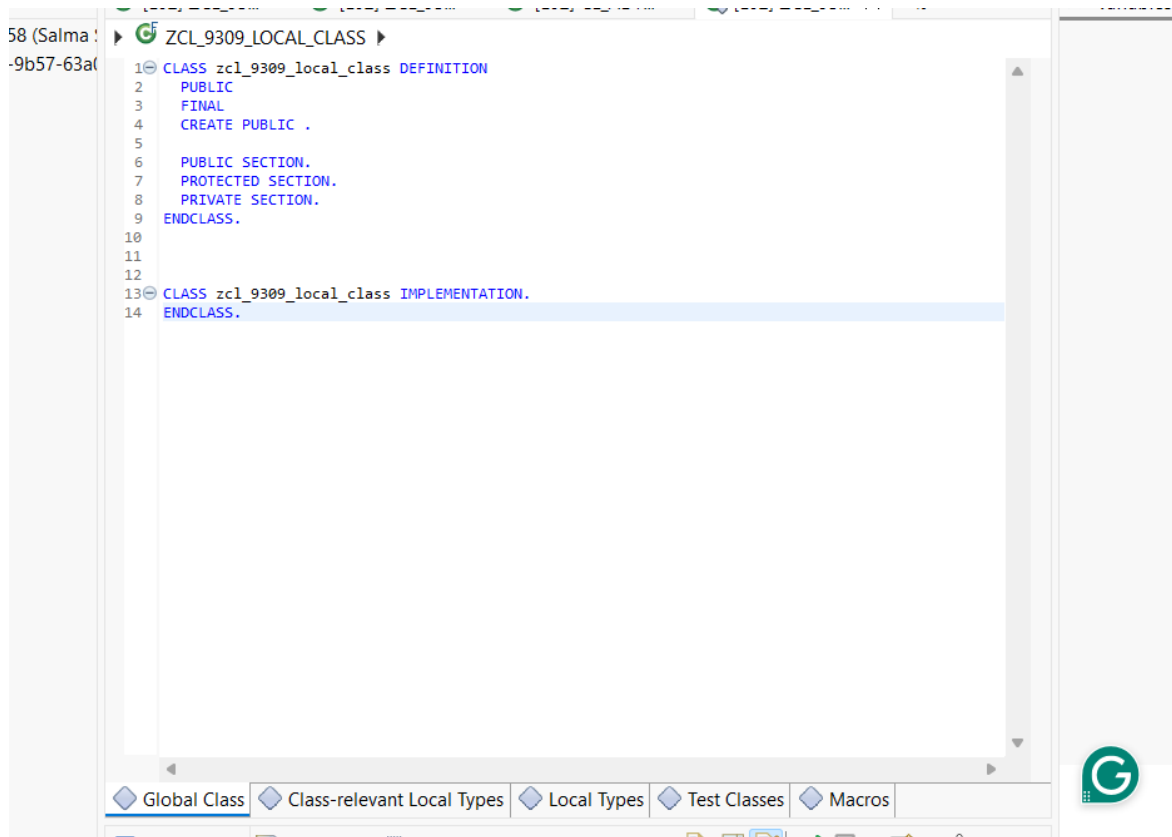
8. Why CREATE PRIVATE Was Removed

Code completion initially generates CREATE PRIVATE. The exercise specifically requires removing it, so the final declaration is CLASS lcl_connection DEFINITION.

9. Activation and Testing

The class was activated with Ctrl + F3. No F9 test is required because the MAIN method does not contain executable logic yet. The ABAP Console can therefore remain empty.

10. Screenshots



Global class ZCL_9309_LOCAL_CLASS created in Eclipse.



ABAP Console visible in Eclipse.

11. Final Checklist

- Global class created ✓
- IF_OO_ADT_CLASSRUN added ✓
- Local class LCL_CONNECTION created ✓
- CREATE PRIVATE removed ✓
- carrier_id added ✓
- connection_id added ✓
- conn_counter added as CLASS-DATA ✓

- Implementation created ✓
- Class activated successfully ✓
- Exercise 8 completed ✓

12. Key Learning

Exercise 8 teaches how to define a local helper class inside a global class and how to distinguish instance attributes (DATA) from a shared class attribute (CLASS-DATA).